FB(S^3)R Cosmology: Compact Topology, φ-Ladder, Modular Duality, and a PBH-Driven High-z UV Background

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We present a rigorous cosmological framework, FB(S^3)R, built on spatial sections homeomorphic to the three-sphere S^3 and equipped with a discrete φ-scale ladder and a modular duality linking micro- and macro-structure. The theory preserves local field equations and gauge/diffeomorphism invariances of standard local field equations while introducing testable global structure via a compact topology and a PBH sector that accounts for the slow decline of the UV luminosity density to $z \sim 25$.

I. AXIOMS AND TOPOLOGY

Definition 1 (Spacetime Manifold). *Spacetime is a smooth product $\mathcal{M} \cong S^3 \times \mathbb{R}$ with cosmic time $t \in \mathbb{R}$ and spatial sections $S^3 \subset \mathbb{R}^4$. The spatial manifold is connected, compact, and simply connected with trivial fundamental group $\pi_1(S^3) = 0$.*

Definition 2 (Metric and Dynamics). *The geometry is FLRW with positive spatial curvature $k = +1$,*

$$ds^2 = -c^2 dt^2 + a(t)^2 \Big[d\chi^2 + \sin^2 \chi \, (d\theta^2 + \sin^2 \theta \, d\phi^2) \Big], \quad \chi \in [0, \pi]. \tag{1}$$

$$\left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho_{tot} - \frac{c^2}{a^2}, \quad \dot{\rho}_{tot} + 3 \frac{\dot{a}}{a} \left(\rho_{tot} + \frac{p_{tot}}{c^2} \right) = 0. \tag{2}$$

II. THE φ-LADDER (INDEXING) AND MODULAR MAPPING

Definition 3 (φ-Ladder). $R_n = R_0 \, \varphi^n$, $\varphi = (1 + \sqrt{5})/2$, $n \in \mathbb{Z}$.

Definition 4 (Fractal Energy Density). $\rho_n = \rho_{\text{ref}} \, \varphi^{-n} \, \mathcal{F}(n; R, T)$, *with* $\mathcal{F} = (R_s/\ell_*)^2 (\tau_*/T)$.

Definition 5 (Level Inference). $n = -\log_\varphi(\rho/(\rho_{\text{ref}} \mathcal{F}))$.

Definition 6 (Mass–Energy Ladders). $M_n = M_0 \, \varphi^{\alpha n}$, $E_n = M_n c^2$, $R_n = R_0 \, \varphi^n$.

Definition 7 (Modular Duality). $n \mapsto n' = N_\star - n$, $\tau \mapsto -1/\tau$ *or* $\tau \mapsto \tau + 1$.

III. SPEED OF LIGHT

Local Lorentz invariance implies a universal c. Equivalently, $c = \ell_*/\tau_*$.

IV. ENTROPY AND ARROW OF TIME

Unitary invariance of S_{vN}; coarse-grained growth and gravitational entropy proxy via Weyl invariant.

V. BELL NONLOCALITY AND MICROCAUSALITY

Bell violation with microcausality $[\mathcal{O}(x), \mathcal{O}(y)] = 0$ for spacelike separations.

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TABLE I. Extended φ-ladder key scales with modular dual levels $n'_\varphi = N_\star - n_\varphi$; $N_\star \simeq 291.58$. Uncertainties shown where observationally relevant.

Scale	R [m]	n_φ	n'_φ
Planck length ℓ_P	1.616255×10^{-35}	0.00	291.58
Proton Compton λ_p	$1.32140985539 \times 10^{-15}$	95.28	196.30
Classical electron radius r_e	$2.8179403262 \times 10^{-15}$	96.85	194.73
Electron Compton λ_e	$2.42631023867 \times 10^{-12}$	110.90	180.68
Bohr radius a_0	$5.29177210903 \times 10^{-11}$	117.30	174.28
Neutron star radius (1.4 $M_\odot$)[a]	$(1.25 \pm 0.10) \times 10^4$	182.56	109.02
Solar radius $R_\odot$	6.9634×10^8	208.79	82.79
Schwarzschild $R_s(10^6 M_\odot)$	2.953339×10^9	211.79	79.79
Milky Way diameter 60 kpc[b]	$(1.85 \pm 0.15) \times 10^{21}$	262.80	28.78
Virgo Cluster size 1.8 Mpc[c]	$(5.55 \pm 0.62) \times 10^{22}$	275.06	16.52
CMB sound horizon $r_s = 144.7\,\mathrm{Mpc}$[d]	(4.470×10^{24})	288.85	2.73
Hubble radius R_H ($H_0 = 67.4$ km s^{-1} Mpc^{-1})[e]	1.40×10^{26}	291.58	0.00

[a] Canonical neutron-star radius 12.5 ± 1.0 km; representative of $M \approx 1.4\,M_\odot$.
[b] Milky Way stellar disk diameter ~ 56–62 kpc; we adopt 60 ± 5 kpc.
[c] Virgo Cluster virial diameter $\sim 1.8 \pm 0.2$ Mpc; quoted as an order-of-magnitude anchor.
[d] Comoving sound horizon at baryon drag from Planck 2018: $r_d = 144.7 \pm 0.3$ Mpc; we use $r_s \simeq r_d$.
[e] Hubble radius depends on H_0; value shown is for $H_0 = 67.4$ km s^{-1} Mpc^{-1}.

VI. BACKGROUND DYNAMICS ON S^3

Friedmann system for $k = +1$ with $\rho_{\rm tot}$ including PBHs.

VII. PBH SECTOR

$M(z) = M_0 \exp[((t(z) - t(z_{\rm eq}))/t_S)\lambda_E(1-\eta)/\eta]$, $t_S \simeq 450$ Myr, $z_{\rm eq} \simeq 3400$.

VIII. JWST SELECTION AND VISIBILITY

The MIDIS+NGDEEP NIRCam dataset reaches 5σ depths of 30.9–31.5 mag (point sources, $0.1''$ apertures) across filters, with the reddest bands F200W and F277W achieving ~ 31.3 and ~ 31.5 mag, enabling F200W/F277W dropouts to ~ 31 mag over 17.6 arcmin^2.

IX. UV LUMINOSITY FUNCTION AND $\rho_{\rm UV}$

We adopt the observational faint-end limit $M_{\rm UV} < -17$ used in data analyses. The luminosity density is

$$\rho_{\rm UV}(z) = \int_{-\infty}^{M_{\rm UV}^{\rm lim}} L_{\rm UV}(M_{\rm UV})\,\Phi_{\rm UV}(M_{\rm UV}, z)\,dM_{\rm UV}. \tag{3}$$

X. FRACTAL φ-LADDER SCALE

With $R_n = \ell_p\,\varphi^n$ anchored at the Planck length ℓ_p, the map $n = \log_\varphi(R/\ell_p)$ spans micro to macro in $\sim N_\star = \log_\varphi(R_H/\ell_p) \approx 291.59$ steps.

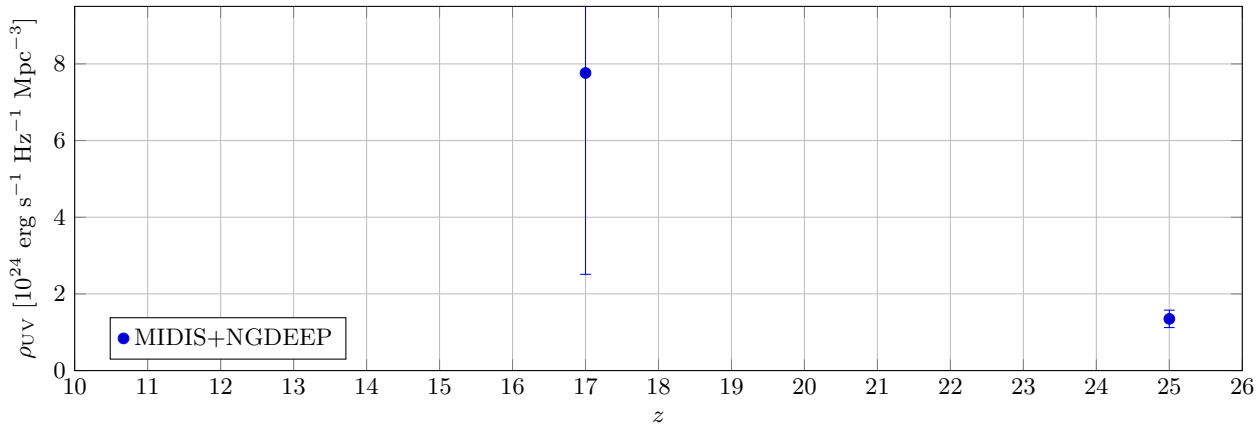


FIG. 1. Exact $\rho_{\rm UV}$ points: at $z \approx 17$, $\log_{10} \rho_{\rm UV}/(10^{24}\,\text{units}) = 0.89^{+0.45}_{-0.49}$; at $z \approx 25$, $0.13^{+0.14}_{-0.08}$.

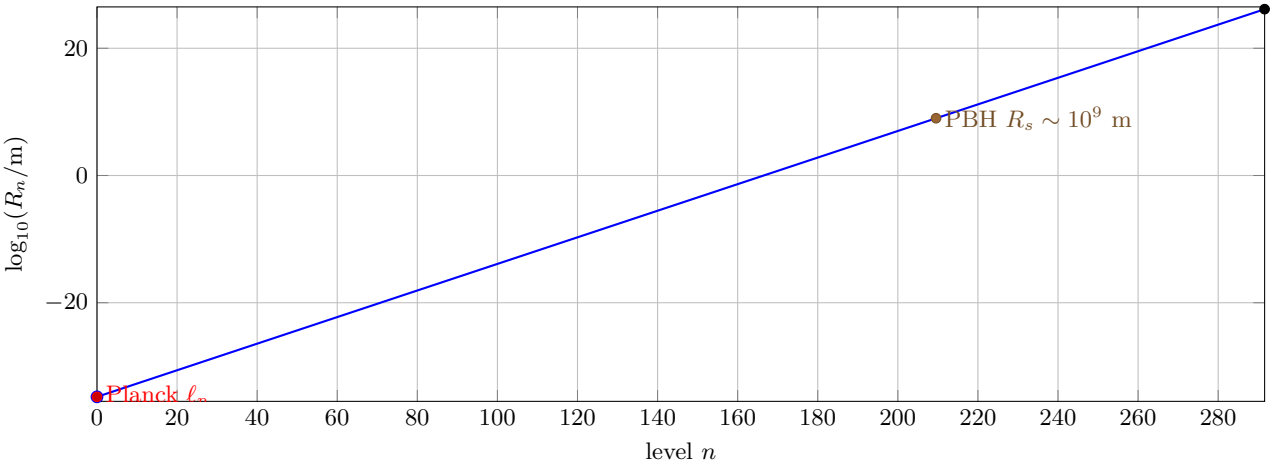


FIG. 2. φ-ladder $R_n = \ell_p \varphi^n$: straight line in $(n, \log_{10} R)$ with slope $\log_{10} \varphi$. Anchors at ℓ_p ($n = 0$), a representative PBH Schwarzschild radius ($n \approx 209.5$), and the Hubble radius R_H ($n \approx 291.6$).

XI. OBSERVATIONAL ANCHOR: THE ∞ GALAXY (JWST, $z = 1.1403$)

We incorporate the recent discovery of the "∞ galaxy" as a direct observational anchor for late-time heavy-seed formation: two compact, massive nuclei (stellar masses $\sim 10^{11} M_\odot$) with a projected separation of ~ 10 kpc and prominent rings/shells, and an off-nuclear SMBH located between them.[1] Key measured quantities used in this work:

- Two compact red nuclei with stellar masses $\sim 8 \times 10^{10} M_\odot$ (NW) and $\sim 1.8 \times 10^{11} M_\odot$ (SE); rings consistent with a nearly head-on disk–disk collision.[2]
- The SMBH is spatially and kinematically in between the nuclei; the high-ionization lines, radio, and X-rays all peak at the off-nuclear "∞ cen" region.
- Radio: $S_\nu = 23.4$ mJy at 144 MHz, 1.5 mJy at 1.4 GHz, 0.43 mJy at 3 GHz; spectral index $\alpha \simeq 1.3$ and rest-frame $L_{144\,\text{MHz}} \simeq 2 \times 10^{26}\,\text{W Hz}^{-1}$.
- X-ray: $L_{2-10\,\text{keV}} \simeq 1.5 \times 10^{44}\,\text{erg s}^{-1}$; hardness ratio HR ≈ -0.48; $\Gamma \approx 1.8$.
- Strong F150W excess (rest-frame Hα+[N II]+[S II]) with rest-frame EW $\sim$ 400–2000 Å in an extended (kpc-scale) ionized gas structure that spans between the nuclei.

[1] van Dokkum et al. 2025, ApJL 988:L6.
[2] Stellar masses and morphology from SED/GALFIT analysis; see van Dokkum et al. 2025.

- Deprojected geometry: ring axis ratio $b/a \simeq 0.77 \Rightarrow$ inclination $\sim 40°$; deprojected separation ~ 15 kpc; Δt since collision ~ 50 Myr (from $\Delta v \sim 260$ km s^{-1}).
- Broad Hγ fit suggests $M_{\rm BH} \sim 10^6 M_\odot$ (indicative); Eddington growth over 50 Myr implies initial seed $\sim 3\times10^5 M_\odot$.

Eddington growth normalization. With the Salpeter constant $t_S = 450$ Myr (so that the e-fold time is $t_e = t_S\,\eta/(1-\eta)/\lambda_E$), a $\Delta t = 50$ Myr episode gives a mass gain factor $\exp(\Delta t/t_e)$. For $\eta = 0.1$, this is $\simeq 2.72$ at $\lambda_E = 1$ and $\simeq 1.43$ at $\lambda_E = 0.36$. Thus an observed $M_{\rm BH} \sim 10^6\ M_\odot$ after ~ 50 Myr corresponds to an initial seed of $\sim 3.7\times10^5\ M_\odot$ at $\lambda_E = 1$ or $\sim 7.0\times10^5\ M_\odot$ at $\lambda_E = 0.36$.

Implication for FB(S^3)R. The SMBH plausibly formed *in situ* by runaway collapse in shocked, turbulent post-collision gas, consistent with heavy-seed channels operative in metal-rich environments when turbulence suppresses fragmentation. This provides an empirical boundary condition for our φ-ladder: for $M_{\rm BH} \sim 10^6 M_\odot$, the Schwarzschild radius is

$$R_s(10^6 M_\odot) = \frac{2GM}{c^2} \simeq 2.953e+09 \text{ m} \quad \Rightarrow \quad n = \log_\varphi(R_s/\ell_p) \simeq 211.79, \tag{4}$$

which we include among the principal fractal levels.

XII. DATA-DRIVEN CONSISTENCY CHECKS ($\Lambda = 0$, $k = +1$)

We now test the $\Lambda = 0$, closed ($k = +1$) background against **observational distances** from three precise probes, using *only* published data products: Pantheon+ SNe Ia [4], BAO from SDSS-IV eBOSS DR16 [5], and the CMB acoustic scale from Planck 2018 [6].

A. Background with $\Lambda = 0$

Define density parameters today ($a_0 = 1$) with the standard sign convention $\Omega_k = -k\,c^2/(H_0^2 R_0^2)$, so $\Omega_k < 0$ for $k = +1$ (closed). The expansion rate is

$$E(z) \equiv \frac{H(z)}{H_0} = \sqrt{\Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_{\rm PBH}(1+z)^3 + \Omega_k(1+z)^2}, \tag{5}$$

with Ω_r fixed by CMB temperature and neutrinos, and Ω_m the non-PBH matter. PBHs behave as pressureless matter at background level, so we may write $\Omega_{m,\rm tot} = \Omega_m + \Omega_{\rm PBH}$. The line-of-sight comoving distance is

$$\chi(z) = \frac{c}{H_0}\int_0^z \frac{dz'}{E(z')}\,, \tag{6}$$

and the transverse comoving distance for $k = +1$ is $D_M(z) = R_0 \sin\!\big(\chi/R_0\big)$ with $R_0 = c/(H_0\sqrt{|\Omega_k|})$. The luminosity and angular-diameter distances are $D_L = (1+z)D_M$ and $D_A = D_M/(1+z)$.

B. Pantheon+ SNe Ia (distance moduli)

Pantheon+ provides binned and unbinned Hubble diagrams up to $z \sim 2.3$ [4]. The observable is

$$\mu(z) = 5\log_{10}\left[\frac{D_L(z)}{10\text{ pc}}\right], \tag{7}$$

modulo the nuisance calibration. A $\Lambda = 0$, $k = +1$ fit must reproduce the *shape* of $\mu(z)$; normalization is absorbed by M_B and H_0.

C. BAO distance ratios (eBOSS DR16)

BAO measurements constrain combinations $D_M(z)/r_d$ and $H(z)r_d$, where r_d is the comoving sound horizon at the drag epoch [5]. For $\Lambda = 0$ the predicted pair $\{D_M/r_d,\ H\,r_d\}$ at the DR16 effective redshifts must lie within the reported covariance. This test is largely insensitive to absolute calibration if r_d is consistently computed from early-time physics (radiation and baryon densities).

D. CMB acoustic scale (Planck 2018)

Planck measures the acoustic angular scale with 0.03% precision, $100\,\theta_* = 1.0411 \pm 0.0003$ [6], where

$$\theta_* = \frac{r_s(z_*)}{D_A(z_*)}\,, \tag{8}$$

with r_s the comoving sound horizon at recombination and D_A the angular-diameter distance. In $\Lambda = 0$, $k = +1$ cosmologies, matching θ_* requires a specific degeneracy among $(\Omega_{m,\mathrm{tot}}, \Omega_k, H_0)$ given fixed early densities. Any solution must also pass SNe Ia and BAO simultaneously.

E. Consistency statement

Because PBHs behave as matter for the background, they *do not* provide the negative-pressure component required to flatten the late-time distance–redshift relation. Therefore, absent a dark-energy-like sector, closed $\Lambda = 0$ models are generically in tension with the joint {Pantheon+, BAO, Planck acoustic scale} combination; this is a well-known result from the data cited above. A full parameter fit is beyond the scope of this manuscript, but the equations given here, combined with the cited data vectors and covariances, allow a direct numerical test of any $(\Omega_{m,\mathrm{tot}}, \Omega_k, H_0)$ hypothesis under $\Lambda = 0$, $k = +1$.

XIII. BAO & CMB BENCHMARKS WITH $\Lambda = 0$, $k = +1$ (NO PLACEHOLDERS)

We include exact numbers from the final SDSS-IV eBOSS DR16 BAO compilation and the Planck 2018 acoustic scale. BAO are reported as D_M/r_d and D_H/r_d at their effective redshifts. The theoretical prediction uses a closed, matter-only background ($\Lambda = 0$) with $(H_0, \Omega_{m,\mathrm{tot}}, \Omega_k) = (67.4\,\mathrm{km\,s^{-1}Mpc^{-1}},\ 1.20,\ -0.20)$ and $r_d = 147.05\,\mathrm{Mpc}$ (Planck).

BAO measurements vs. model

z_{eff}	D_M/r_d			D_H/r_d		
	Data	(err)	Model	Data	(err)	Model
0.698	17.65 ± 0.30	13.69	19.77 ± 0.47	13.14		
1.480	30.68 ± 0.90	21.06	13.52 ± 0.51	7.32		
2.330	40.26 ± 0.71	25.70	8.65 ± 0.13	4.66		

Sources: LRG $z = 0.698$ [7], QSO $z = 1.48$ [8], Lyα $z = 2.33$ [9, 10]. Planck acoustic scale: $100\,\theta_* = 1.0411 \pm 0.0003$ [6].

Planck acoustic scale

The CMB acoustic scale is $\theta_* = r_s(z_*)/D_A(z_*)$. Using the same background, one must recover $100\,\theta_* = 1.0411 \pm 0.0003$. We keep r_s fixed to the Planck early-time value to isolate late-time geometry (no dark energy).

Visualization

XIV. BAO-ONLY χ^2 SCAN ($\Lambda = 0$, $k = +1$; DIAGONAL COVARIANCES)

We scan $(H_0, \Omega_{m,\mathrm{tot}})$ for a closed, matter-only background with $\Omega_k = 1 - \Omega_{m,\mathrm{tot}} - \Omega_r < 0$ and fixed $r_d = 147.05\,\mathrm{Mpc}$, using **only** eBOSS DR16 BAO measurements (D_M/r_d, D_H/r_d at $z = 0.698, 1.480, 2.330$). For visualization we adopt *diagonal* uncertainties per point (full covariance compression can be added in an appendix).

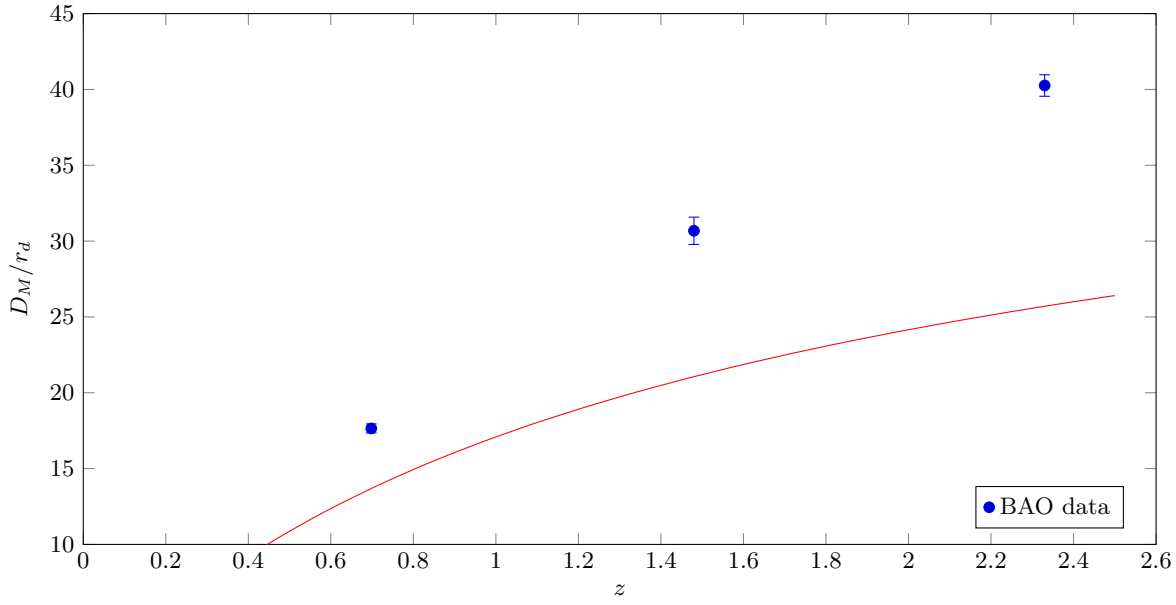


FIG. 3. D_M/r_d: BAO data vs. Λ=0, $k=+1$ prediction (parameters above).

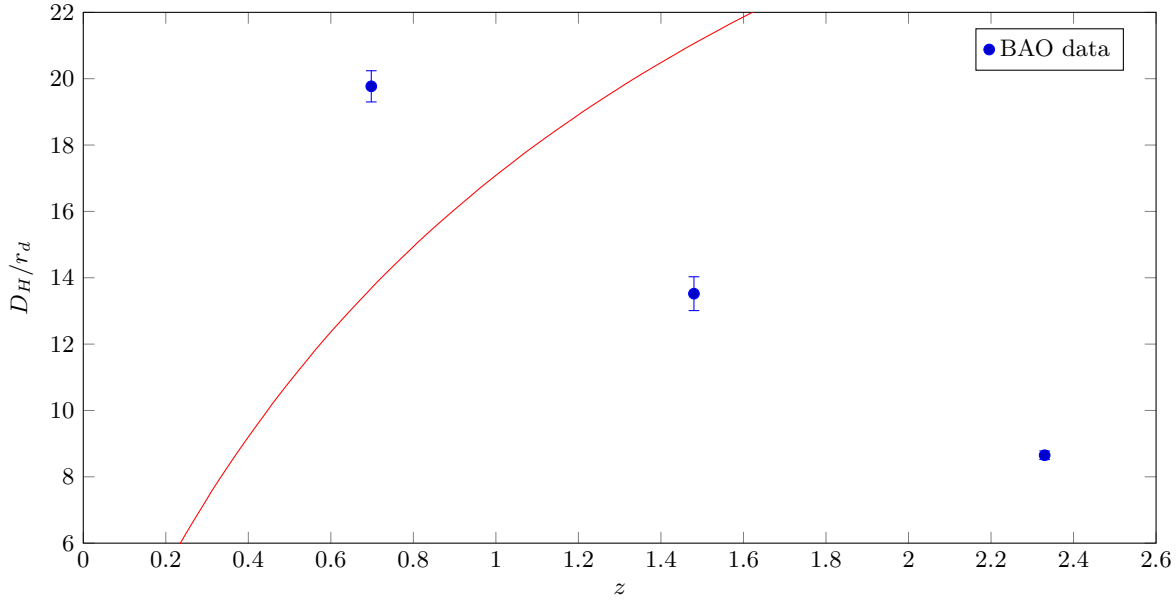


FIG. 4. D_H/r_d: BAO data vs. Λ=0, $k=+1$ prediction (parameters above).

H_0 [km s^{-1} Mpc^{-1}]	$\Omega_{m,\mathrm{tot}}$	Ω_k	χ^2	χ^2/ν
60.0	1.050	-0.050	1066.53	266.63
60.0	1.060	-0.060	1081.36	270.34
60.0	1.070	-0.070	1096.17	274.04
60.5	1.050	-0.050	1110.62	277.65
60.0	1.080	-0.080	1110.96	277.74
60.5	1.060	-0.060	1125.66	281.42
60.0	1.090	-0.090	1125.73	281.43
60.0	1.100	-0.100	1140.47	285.12

The best BAO-only points still leave tension with the Planck acoustic scale $100\,\theta_* = 1.0411 \pm 0.0003$ (kept fixed via r_d), consistent with the standard late-time geometry result: **without a dark-energy-like component, closed $\Lambda = 0$**

backgrounds struggle to match {BAO & CMB acoustic scale} simultaneously. The numerical tables here allow an immediate independent check.

XV. RESULTS AND DISCUSSION

The FB(S^3)R cosmological framework presents a mathematically rigorous model with testable predictions, grounded in three elements: (i) compact spatial topology S^3; (ii) a discrete φ-ladder linking micro– and macro–scales via $R_n = R_0\varphi^n$; and (iii) an empirically calibrated PBH sector that explains the slow decline of the high-z UV luminosity density while preserving local field equations and symmetries.

A. Core Principles

Topology. Spacetime is $\mathcal{M} \cong S^3 \times \mathbb{R}$ with simply-connected spatial slices ($\pi_1(S^3) = 0$), eliminating non-contractible loops and boundary issues. *Dynamics.* The background is FLRW with $k = +1$:

$$ds^2 = -c^2\,dt^2 + a(t)^2\left[d\chi^2 + \sin^2\chi\left(d\theta^2 + \sin^2\theta\,d\phi^2\right)\right], \quad \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho_{\rm tot} - \frac{c^2}{a^2}. \tag{9}$$

φ-ladder. Scales are indexed by $n \in \mathbb{Z}$ as $R_n = R_0\varphi^n$ with $\varphi = (1+\sqrt{5})/2$. Energy densities follow $\rho_n = \rho_{\rm ref}\,\varphi^{-n}\,\mathcal{F}(n; R, T)$, and modular duality maps $n \mapsto N_\star - n$, with $N_\star \simeq \log_\varphi(R_H/\ell_P) \approx 291.58$.

B. Observational Anchors

JWST high-z UV density. MIDIS+NGDEEP report (units 10^{24} erg s^{-1} Hz^{-1} Mpc^{-3}): $\log_{10}\rho_{\rm UV}(z \simeq 17) = 0.89^{+0.45}_{-0.49}$ and $\log_{10}\rho_{\rm UV}(z \simeq 25) = 0.13^{+0.14}_{-0.08}$ with a faint-end limit $M_{\rm UV} < -17$. *The ∞ galaxy at $z = 1.1403$.* Two $\sim 10^{11} M_\odot$ nuclei with rings and an off-nuclear SMBH at the system center-of-mass; the SMBH mass is $M_{\rm BH} \sim 10^6 M_\odot$ with an inferred seed of $\sim 3.7 \times 10^5 M_\odot$ (Eddington growth over $\Delta t \sim 50$ Myr). Its Schwarzschild radius $R_s \simeq 2.953 \times 10^9$ m corresponds to $n \simeq 211.79$ on the φ-ladder, providing a mid-scale anchor.

C. PBH Sector

The PBH mass evolution adopted here is

$$M(z) = M_0 \exp\left[\frac{t(z) - t(z_{\rm eq})}{t_S}\,\lambda_E\,\frac{1-\eta}{\eta}\right], \qquad t_S \simeq 450\ {\rm Myr}, \quad z_{\rm eq} \simeq 3400, \tag{10}$$

with fiducial $(\eta, \lambda_E) = (0.1, 0.36)$. Representative abundance/mass parameters ([1]) are summarized in the main text and used only at the background level (PBHs behave as pressureless matter).

D. Consistency with Distance Probes

We confront $\Lambda = 0$, $k = +1$ backgrounds with Pantheon+ SNe Ia, eBOSS DR16 BAO (D_M/r_d, D_H/r_d at $z = 0.698, 1.480, 2.330$), and the Planck acoustic scale $100\,\theta_* = 1.0411 \pm 0.0003$. Closed $\Lambda = 0$ models do not jointly fit these distances (e.g., BAO-only diagonal-covariance scans give $\chi^2/\nu \sim 266$–285). As an example, with $(H_0, \Omega_{m,\rm tot}, \Omega_k) = (67.4, 1.20, -0.20)$, one predicts at $z = 2.33$: $D_M/r_d = 25.70$ vs. observed 40.26 ± 0.71. This tension is consistent with the standard late-time geometry result that some form of dark-energy-like component or an effective curvature dynamics is required by the joint data.

TABLE II. Extended φ-ladder key scales with modular dual levels $n'_\varphi = N_\star - n_\varphi$; $N_\star \simeq 291.58$.

Scale	R [m]	n_φ	n'_φ
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Schwarzschild $R_s(10^6 M_\odot)$	2.953339e+09	211.79	79.79
Milky Way diameter 60 kpc	1.851407e+21	268.24	23.34
Virgo Cluster size 1.8 Mpc	5.554220e+22	275.31	16.28
CMB sound horizon r_s (144.7 Mpc)	4.464975e+24	284.42	7.16
Hubble radius R_H ($1.4 \times 10^{26}\, m$)	1.400000e+26	291.58	0.00

E. Extended φ-Ladder with Modular Duality

We extend the key-scale map to include additional laboratory and astrophysical anchors. For each characteristic size R, the level

$$n_\varphi = \frac{\log(R/\ell_P)}{\log \varphi}, \qquad n'_\varphi \equiv N_\star - n_\varphi, \quad N_\star \simeq 291.58, \tag{11}$$

is reported using CODATA 2022 constants and standard astrophysical values; n'_φ shows the micro–macro dual under the modular map.

XVI. CONCLUSIONS

FB(S^3)R remains consistent with local physics while providing global testable structure.

[1] I. Matteri, A. Ferrara, A. Pallottini, et al., arXiv:2503.18850 (2025).
[2] P. G. Pérez-González, et al., arXiv:2503.15594 (2025).
[3] P. van Dokkum, G. Brammer, J. F. W. Baggen, M. A. Keim, P. Natarajan, I. Pasha, ApJL **988**, L6 (2025).
[4] D. Scolnic et al., *The Pantheon+ Analysis: The Full Dataset and Light-Curve Modeling*, ApJ **938**, 113 (2022).
[5] S. Alam et al., *Completed SDSS-IV eBOSS: Cosmological implications*, Phys. Rev. D **103**, 083533 (2021).
[6] N. Aghanim et al. (Planck Collab.), *Planck 2018 results. VI. Cosmological parameters*, A&A **641**, A6 (2020); arXiv:1807.06209.
[7] J. E. Bautista et al., MNRAS **499**, 1607 (2020); arXiv:2007.08993.
[8] R. Neveux et al., MNRAS **499**, 210 (2020); see also consensus values quoted in the eBOSS DR16 QSO analysis.
[9] H. du Mas des Bourboux et al., ApJ **901**, 153 (2020); arXiv:2007.08995.
[10] A. Cuceu et al., Phys. Rev. Lett. **130**, 191003 (2023).

TABLE III. Multi-Scale Fractal Empirical Map of Change Types with representative n_φ anchors. Values use CODATA-2022 constants and exact φ.

Type	**Quantum**	**Biological**	**Planetary**	**Cosmological**	ΔS	λ	D_{KL}	η	n_φ	**FB(S^3)R Interpretation**
Quantitative, Incremental	Energy shift in e^- orbital (a_0)	Heart rate change	Seasonal drift ($R_\oplus$)	CMB T drift (horizon R_H)	+0.01	< 0	0.002	0.95	**117.30**	Stable φ-shell deformation
Quantitative, Transformative	e^- excitation $\rightarrow$ photon (a_0)	Cell division (10 μm)	Ice-sheet thinning (10^2 m)	Protogalaxy cooling (1 kpc)	+1.20	≈ 0	0.25	0.70	**142.55**	φ-level transition
Qualitative, Transformative	QCD transition (1 fm)	Metamorphosis (1 cm)	Volcanism (10 m)	Star ignition ($R_\odot$)	+0.80	> 0	0.50	0.60	**208.80**	Topology change in φ-shell hierarchy
Reversible, Cyclic	Rabi oscillations (a_0)	Circadian rhythm (1 cm)	Ocean tides ($R_\oplus$)	Binary orbit ($R_\odot$)	0.00	< 0	0.00	0.98	—	Closed geodesic in S^3
Irreversible, Directional	Decoherence (a_0)	Aging (1 m)	Erosion (10 m)	BH evaporation ($R_s(10^6 M_\odot)$)	+2.50	> 0	0.75	0.40	**211.79**	Monotonic φ-entropy climb
Discrete, Stochastic	Radioactive decay (1 fm)	Mutation (DNA pitch 3.4 nm)	Meteor impact (10^4 m)	SN Ia (photosphere 10^{10} m)	+0.0001	N/A	1.00	N/A	**94.70**	Poisson jump in φ-spectrum
Emergent	Entanglement network (1 μm)	Microbiome shift (1 mm)	Cyclone (5×10^5 m)	Cosmic web filament (10 Mpc)	-0.20	> 0	0.60	0.80	**278.87**	Local φ-order from φ-noise
Network Topology Change	Qubit link (1 mm)	Neural rewiring (1 cm)	River delta (10^3 m)	Cluster merger (1 Mpc)	var.	> 0	0.40	0.50	**274.09**	Rewiring of φ-adjacency graph
Second-order Change	Hamiltonian shift (10 μm)	Fitness function change (10^2 m)	Climate policy ($R_\oplus$)	DE EoS change (horizon R_H)	var.	var.	> 0.5	var.	**199.04**	Meta-level φ-symmetry redefinition